Example: Which customers in each product spend more than their own product's average

Step 1
product CTE
Avg spend
per product

Step 2
Above_average CTE
Join + filter
> Avg

Step 3
Final select
Top 10 results

```
With product_averages As (
Select product, AVG (Total_spend) as AVG_spend
from customer
Group by product
),
Above_average as (
Select A.Name, A.product, A.Spend as above_by
From Customer A
Join above_average B on A.product = B.product
Where A.Total_spend > B.Avg_spend
)
Select *
from above_average
Order by above_by desc
limit 10;
```

Exists, Not Exists, Any, ALL

Exists - Does this subquery return at least one row
No Exists - True when zero rows come back

Example: Customer who placed at least one order

For each customer row

Select 1 from fact_orders of
where f.customer_ID = A.customer_ID

found ≥ 1 row
True → includes row

found 0 rows
False → Excluding row

Fact_orders

OrderID	customer_ID	revenue	Quantity
1	2	100	3
2	3	200	2
3	2	20	1
4	1	50	2

Dim_ customer

Custemer ID	Name	Address
1	Jaben	New road
2	Jane	1st Ave
3	Lerako	Vilakazi
4	Thabo	New str

```
Select
A. Customer_ID
A. Name
From Dim_customers A
Where Exists ( Select 1
               From fact orders f
               Wher A. customer ID = F customer ID
               ;
```

Customers who never ordered

```
Select * customer_id , * Name
From Dim_Customers as A
Where( Select 1
    From Fact_orders as F
Where A.customer_id = F.customer_id
) ;
```

Any and All

True if atleast one value satisfies the condition

All- True only if every single value satisfies it

Example : products cheaper than Every gold purchase

```
Select product_name, price
From dim_Products as A
Where price < Any ( select unit_Price
    From fact_orders F
Join dim_customers C on
F.customer ID = C.customer
Where C.tier = 'Gold'
) ;
```

List of unit price returned by Subquery

[5.00 ; 10 ; 20 ; 7.50]

Dim_Products

Product ID	Product Name	Price	< ALL	< Any
1	Coffee	10	F	T
2	Tea	7.50	F	T
3	Chocolate	15	F	T

views and stored procedure

create table product

as

(select ---

) ;

View → Do not store data

- Similar to CTAS

Create view vw_customer_order_Summary

as

(

) ;

* Use it exactly like a table

Select *

From VW_customer_order_Summary

Create or replace view VW_customer_order_summary

as(

);

* Remove the view

Drop view if exists VW_customer_order_summary ;

Show view ; - list views in your database

Stored ~~proc~~ procedure

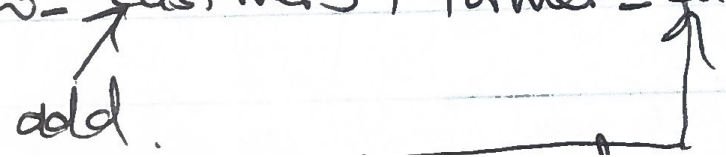
- A named block of SQL logic

Saved in the database that you can execute on demand

- Use a call statement to execute stored proc

- Insert, update, delete

- Tables : New_customers, former_customers, update_cust



```
If (Delete account)
Delete from New_customers
Where customerID = '123';

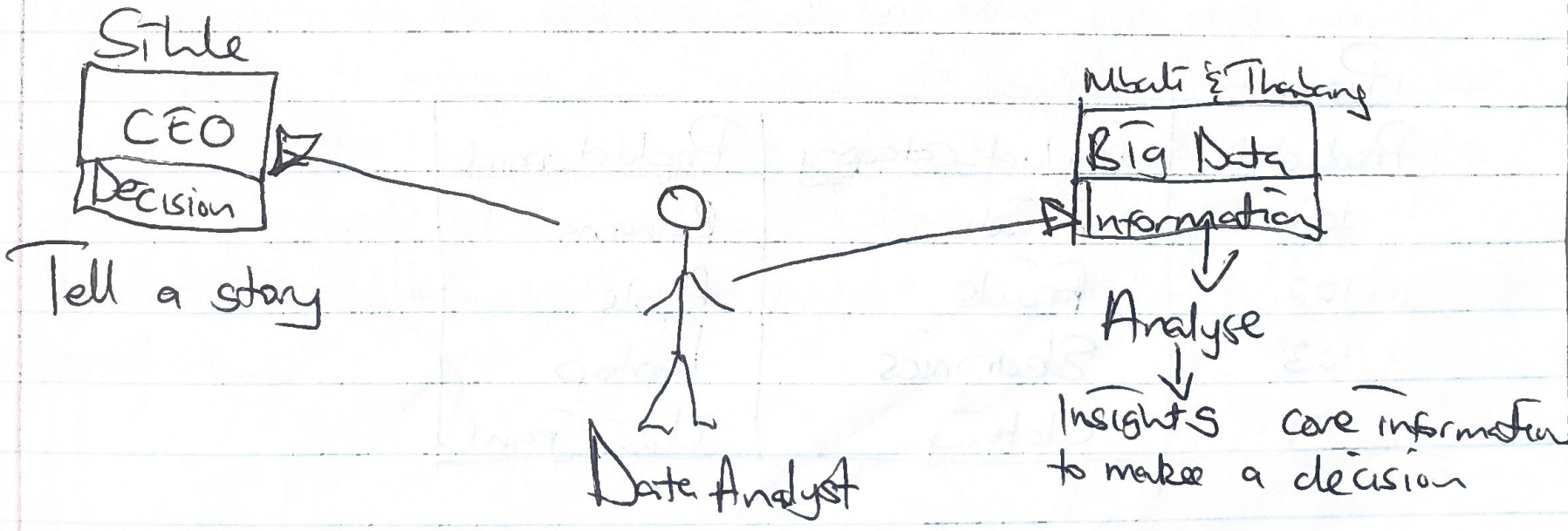
Insert into former_customers
Where customerID = '123';

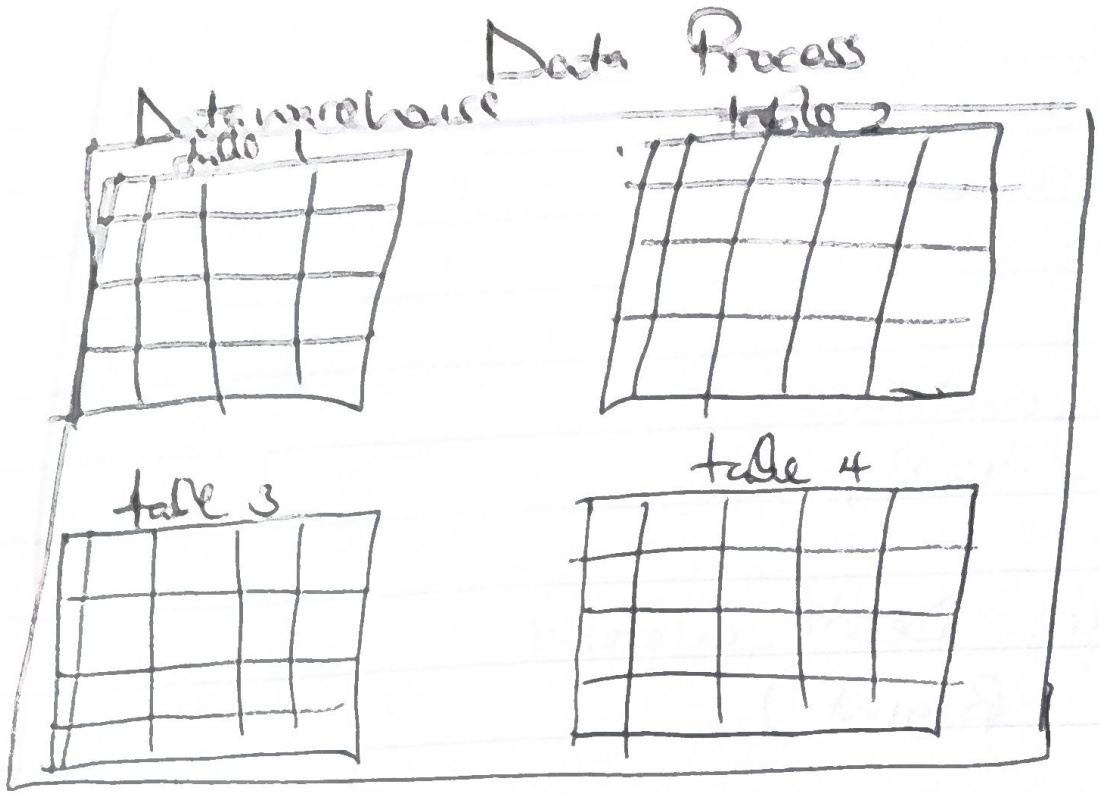
Create procedure Delete_customer
In customerID Bigint)
Begin
If P.customerID in ( select customerID
                     From new_customers
                     )
end if;
```

Data Visualization

Data $\neq$ Is Information

- You need to be able to tell a story from your data
- You need to drive or extract actionable insights from your data
- Actionable insights is to drive Business Decisions





Relational database is structured data that can be related joined together to become.

- In a relational database we store structured data that have (dimension tables & fact tables) that can be connected to one another using primary key and foreign key

Customers

customerID	Name	Age	City
1	Rofhi	28	NY
2	Jeff	22	LA
3	Mark	18	MG

transactions

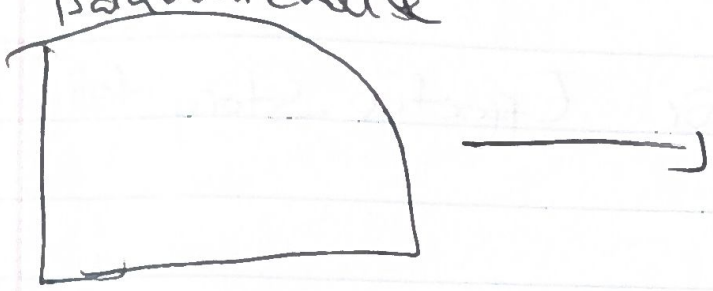
order	cust_ID	ProductID	Re
324	2	104	70
333	1	102	80
401	3	101	100

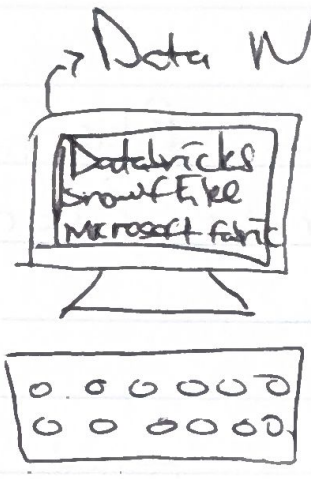
Products

ProductID	Product_category	Product_name
101	Fruits	Banana
102	Fruits	Apple
123	Electronics	laptop
104	Clothing	Uzzi jeans

- Dim table - Dimension table contains descriptive information. A Dim table does not constantly change.
- Primary key is a unique identifier that differentiate each component from the table
- Fact table - has foreign numbers
- Used for analysis.
- Foreign key - is a primary key from one table that is found in another table.

Datawarehouse

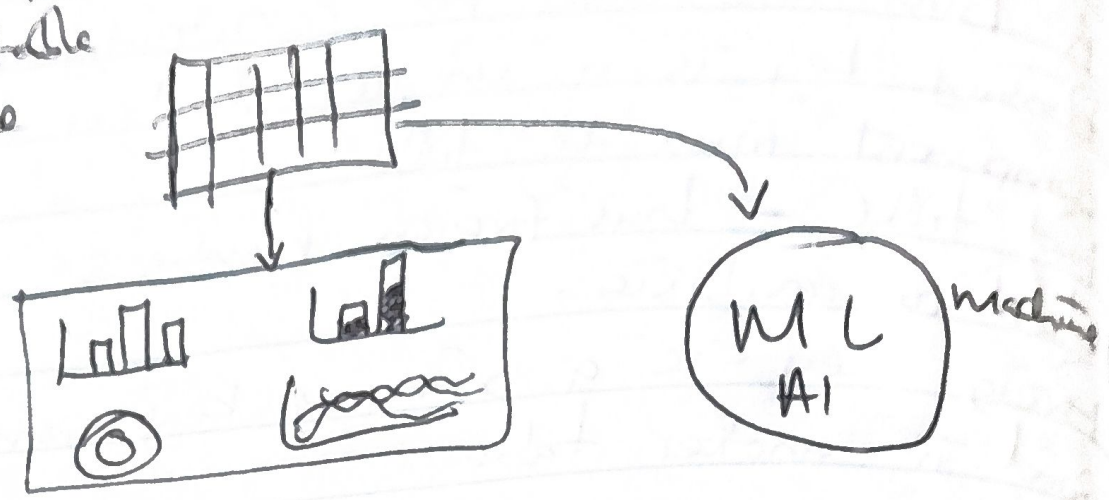




In data management system (DMS) we that where we do the coding
We use it to access the data.

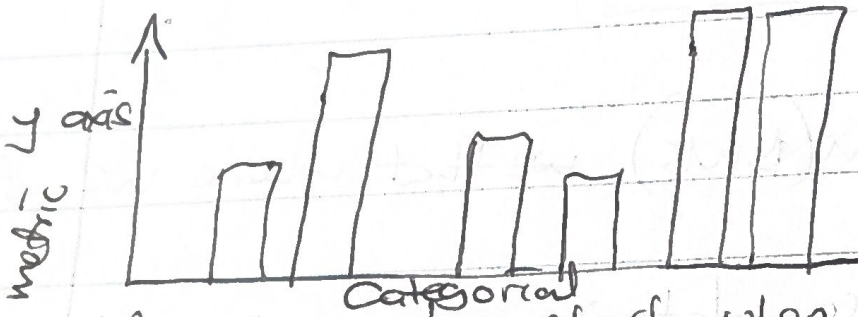
Why do we need to code?
Data inspection
To see what data is available
Data exploratory analysis
To understand the relationship between our data points
- We code to retrieve, to amend, to update, to manipulate, to clean, to aggregate, to join, to delete, to update with the goal of extracting actionable insights
- Feature Building - creating feature used in ML for predictions

- Summarized table
- Aggregated table
- Combined table
Unified table



06/06/26 The Data Design Blueprint
- Choosing the right chart for Effective Story telling

① Bar chart



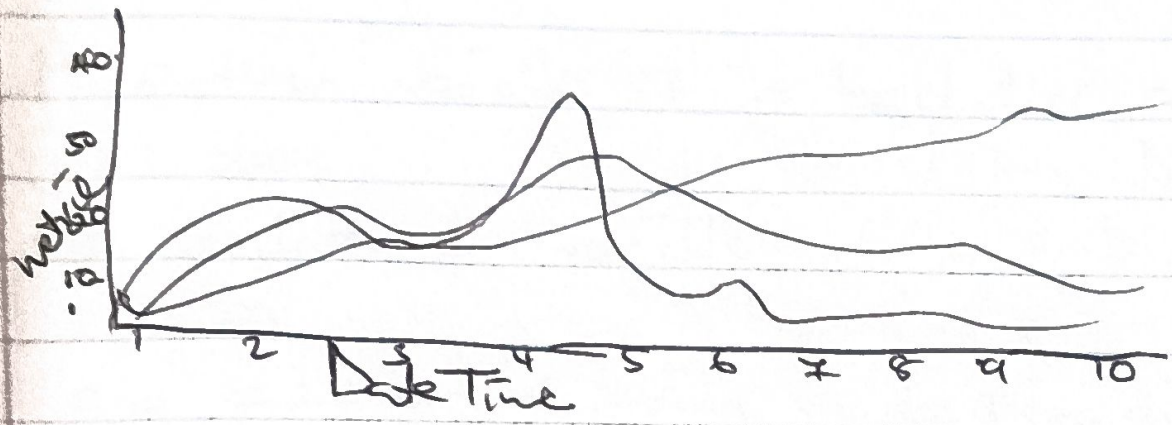
- We use a bar chart when comparing categories

Metric is something you are measuring
Metric is numerical data or quantitative data.

Categorical data is grouping data or is data that puts our information into groups & mainly words
e.g gender; male, female
Qualitative data

A split / legend gives you a detailed breakdown
granularize

Line chart

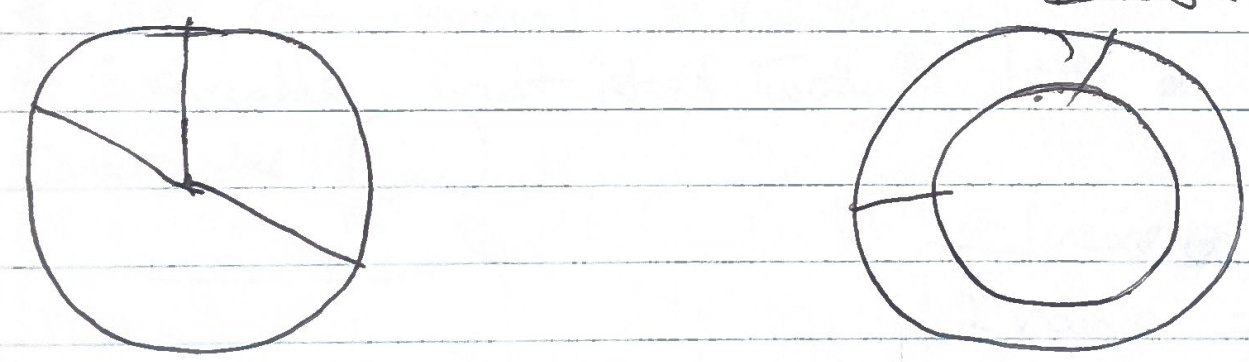


Line chart shows trends over time

Date Time
Month
Day of week
year
Date

} Continual trends

3) Pie chart (showing proportions)



I want to see the contribution of which category generated more revenue

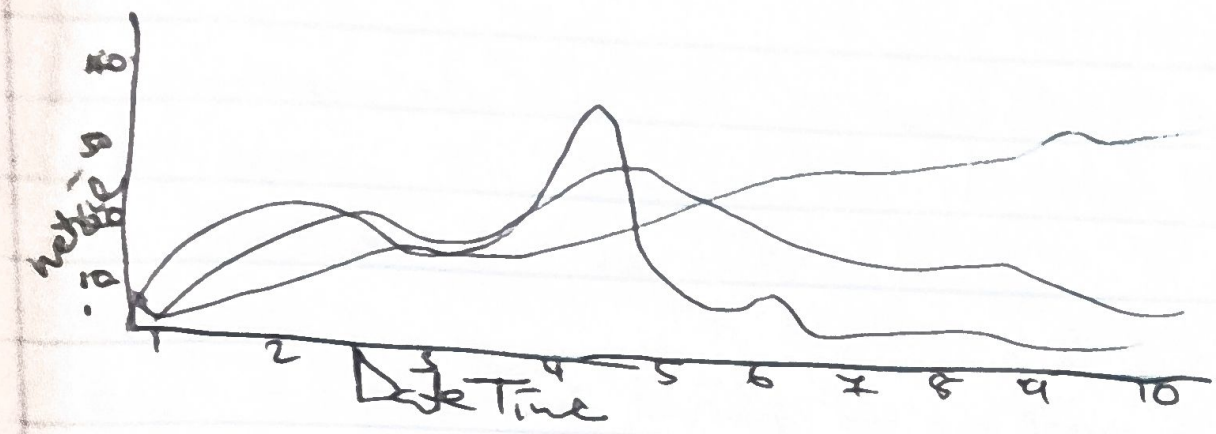
- Categorical Breakdown
- Metric to proportion

4) Stacked chart



Showing proportions for two different categorical data by a specific metric

Line chart

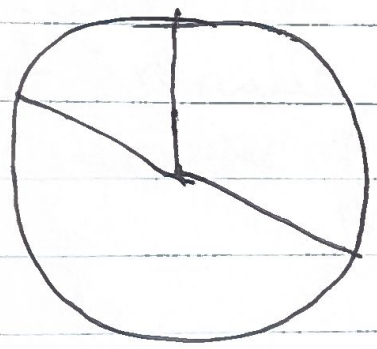


Line chart shows trends over time

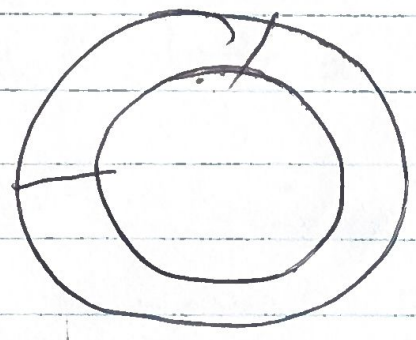
Date Time
Month
Day of week
year
} Continued trends

Date

3. Pie chart (showing proportions)



Donut



I want to see the contribution of which category generated more revenue

- Categorical Breakdown
- Metric to proportion

4. Stacked chart



Showing proportions of two different categorical data by a specific metric

Python fundamentals

A python variable is a bucket or container that store your data

- We declare a variable

x = 20

age = 20 in this variable we are storing a int-integer (number without a decimal)

Name = "Rofhiwa"
Car = "Suzuki" } this are strings — a combination of characters. We have double quotes because it's a string

Print ( ) is a function in python that display what is inside

x = 5
Print (x)

Surnames = ("Nemukula", "Dagada", "Sithole", "Sisiya")
Print (surnames)

Rules of naming variable
A variable must start with a letter or an underscore character (_)

	Incorrect variables
name	2name = "Rofhi"
_name	name surname = "Rofhi"
_name_Surname	name> Surname
name10	

- You can not start a variable name with a number
- You cannot have a space on the name of the variable
- You can not separate the words in a variable name with a dash (-) rather use an underscore (_)

You can declare a variable by starting with an un

9/6/26 Python fundamental

- Variables are case-sensitive

Gender = 'Male'
gender = 'female'
GENDER = 'Gay'

These are three different variable names

A variable name cannot be a python keyword or func
Use variable names that are descriptive and are no
def import python functions
A variable name should be ~~diff~~ defined based on the kind of data that it holds

Ways of Variable Naming

① Camel case
- Each word, expect the first letter start with a capit letter

~~1.~~ myCarName = "Suzuki"

② Pascal Case
- Each word starts with a capital letter

MyNameSurname = "Rofhi Nom"

③ Snake Case
- Seperate each word by underscore

my_name_surname = Rofhi_Nom

Python Data Types

1. Int -> number without a decimal / are whole numbers +
2. Float -> numbe with a decimal +
3. Boolean -> True or false (switch ON or OFF)
4. Datetime -> is just any date.
   "YYYY-MM-DD"
5. String/Object -> combination of characters

10/06/26
Python data type
Text Type: str
Numeric Type: int, float, complex
Sequence Type: list, Tuple range
Mapping Type: dict (dictionary)
Boolean Type: bool

dom or kap
df is the name of the variable
- A string/str is a combination of characters and we use single or double quotes when we have a string
- When we have a multiline string - we use 3 single quotes at the start and end of the string

Functions Applied to Strings
- len() - to count the number of words/characters in a string. and the space in between "Primrose Mukweta" 17
- Upper() - change words in a string to be upper case
- Lower() - change words in a string to be lower case
- Strip() - remove unnecessary empty space in a string

- Split() - seperate words by a comma

On the name of a variable, we do not put single or double quotes

```
name_surname = "Rofhiwa Nemukuta"
len(name_surname)
16
```

Upper() = Changes the values in a variable into upper case

```
print(name_surname.upper())
```

ROFHIWA NEMUKUTA

lower() = changes the values in a variable into lower case

```
print(name_surname.lower())
```

rofhiwa nemukuta.

```
mxit_name = "Rofhiwa"

print(mxit_name.upper())
print(mxit_name.lower())
```

15/06/26.

Mapping type:

What is a dictionary in Python.

- It is used to store key : value pairs

left-side you have the key and on the right-side yo

```
my_profile = {
"name" : "Primrose",
"Surname" : Mukwedeka,
Age : 31,
"Location" : "USA"
}
```

A dictionary is changeable
A dictionary is ordered
A dictionary does not allow duplicates of keys
A dictionary has curly brackets
A dictionary can contain multiple data types

Duplicates in a Dictionary
- In a dictionary the value can be duplicated but not the "key"
- The "key" of the dictionary are sensative ~~case~~
- The length LEN() of the count of the number of key value pairs

Len(my_profile)

Accessing value from dictionary

```
my_car = {
"name": "Suzuki",
"model": Baleno,
"year": 2023,
"dealership": "Soweto"
```

You can access values from a dictionary by using the key

```
print(my_car ["name"])
print(my_car.get("model"))
```

This gives you the keys in your dictionary

```
my_car.values()
```

16/06/25

This is a way of removing a key in a dictionary

my_car

Pop deletes or removes

car_details.pop("dealership")
print(car_details)
~~# delete~~

Update dictionary

17/06/26

my_profile = {
key value
"name" : Rofhiwa
"Surname" : Nemukula
"age" : 28
}

my_profile = dict(
name = 'Rofhiwa',
Surname = "Nemukula",
age = 28,
) Location = "USA"

my-name
my_profile['Name']
Primrose

Functions used in accessing values from a dictionary
key() - Allows to access all key name
Value() - Allows to access all the values in dictionary
items() - put your key: value pairs in a list
get() - Extract a value by specifying the key

my_profile["name"]
Primrose

my_profile.get("name")
Primrose

How to remove values from a dictionary
pop() => my_profile.pop('name')
popitem()
del

pop() => It will delete the name from the dictionary
popitem() - deletes the last key: value pair in the dictionary
.clear() - empties the dictionary

List
- is an array used to store multiple values in a single value variable.
- my_cars = ["BMW", "Polo", "Suzuki"]
- Square brackets
- Indexed

22/06/24
Logical operators
We can use the logical operators to build conditions for data following as an example
== , != , < , > , <= , >=

```
x = 10
y = 20

if x < y:
    print("Correct")
```

```
x = 50
y = 20
if x<y:
    print("correct")
elif x>y:
    print("Nothing")
else:
    print("Brightlearn")
```

Python functions
A block to code that runs when called.
Avoid code repetition
Cuts down the size of a file
makes the code to be more organised.
A function is re-usable.
Build for a specific task.

```
def greetings():
    print("How are you Rofhiwa")

greeting()

#def means "define a function"
#greetings is the name of the function
# () holds inputs/parameters
#: starts function
#Indented code belongs to a function


#name of a parameter

greet(name)
print(f"Hi {name}")

greet(Ptim)


time = 8
name = Rofhiwa"
if time < 8:
    print(f"Good day {name}")
elif time >= 8:
    print(f"Good Evening {name}")
else
    print("Dear Valued Customer")
```

3/06/24 Pandas

Google colab

→ It's a python library that we use for data processing
→ We use it to analyze and manipulate data.

Importing Libraries

```
import pandas as pd # Data Manipulation
import numpy as np # mathematical calculations
```

Data Ingestion

~~data_path~~

df.columns

columns: shows all the column names that we have on the datsets:

#df.dtypes tells us ~~that~~ the data type in each and every column.

Int - is a whole number, number without decimals

Object - is a string which is a combination of characters

#

-> Shows all the information from the dataset:

~~df.ins dataframe()~~ df.info()

-> Shows us the null values in our dataset

df.isnull()

-> Calculate the number of the null values in each column

~~df.isf~~ df.isnull().sum

df.shape

Shows the number of rows and the number of columns

df

df ["Admission Type"]. unique
~~Shows the number of rows~~
#Checking the different values in a column.
df ["Admission Type"]. unique ()

print (df ["Gender"]. unique ())

Kaggle / Medium / sk learn

24/06/26

Checking the different values in a column.
print (df ["Gender"]. unique ())

Lambda ==>

#Is to select unique values in a specific
df (Admission type"]. unique()
array ('elective', 'Emergency') , dtype = object)

#Counts the number of time each category appears in the data - It also gives you the different value in a column.
df ["Gender"]. value_counts ()

#This line of code is to check for duplicated row in my data.
total_duplicate = df. duplicated (). sum ()
print (total_duplicates)

This code is to check the number of unique patient in my data
df['PatientID'].nunique()

This is how to filter the data based on a specific column:
df[df['Age'] > 70]

Yutube video
import pandas as pd
df = pd.read_csv('')
df

& Checking rows & colums:
df.shape

& Check rows & colums and data types
df.info()

& pd
pd.set_option('display.max_columns'

-> The meaning of each column:
Schema_df:

& Check the first 5 rows
df.head(5)

Check the last 5 rows
df.tail(5)

Lesson 13
Python D. Visualization
Different Libraries to draw the graphs
1. Plotly express
2. Seaborn
3. Matplotlib

1. Matplotlib
It allows to create a wide variety of visualizations including line plots, scatter plots, bar charts, histograms, pie charts and more.

```
Import matplotlib.pyplot as plt

# Data
x = [1,2,3,4,5]
y = [2,4,6,8,10]

# create a line plot
plt.plot(x,y)

# create a scatter plot
plt.scatter(x,y)

# Add labels & title
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Simple Line Plot")
plt

# Display the plot
plt.show()

plt.xticks(rotation=75)

plt.plot(df['Date'], df['close'],
    Label = 'AAPL')
```

(8) Plotly Express

Is a high. level data visualization library use in Python used to create interactive charts. It allows users to zoom, pan, hover over data points and save the visualization

Import plotly.express as px

- fig = px.bar(df, x='Date', y='close', title='Interactive BC')
fig.show()

- fig = px.scatter_3d(df, x='Date', y='close', Z='volume',
title='Interactive 3D Scatter plot')
fig.show()

15/07/24

Evaluation Metric

Model Evaluation

Algorithm used to learn the data pattern & make

Model Evaluation -> is the process of using different evaluation metric to understand a machine learning model's performance as well as the strength and weakness

Importance of Model Evaluation

- You what to test if the model is reliable
- You want to know if the model is capable of making accurate predictions on new and unseen data

Supervised Machine learning

① Classification
Yes / No

② Regression
One value

Classification Metrics

① Confusion Matrix is a table that is used to describe the performance of a classification model on a set of data which the true values are known

	Actual values: Positive (1)	Negative (0)
(1) Positive	True Positive ①	false Positive ③
(0) Negative	false Negative ④	True Negative ②

predicted Values

① Predicted : Positive and Actual value is Positive (TP)
② Predicted : Negative and Actual value : Negative (TN)
③ Predicted : Negative & Actual value : Positive (FN)
④ Predicted : Positive and Actual value : Negative (FP)

② Accuracy is the number of correct predictions over the total predictions

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{total number of predictions}}$$